RONIC OS • CLOUD BACKEND, AUTH & MONETIZATION SPECIFICATION

SUPABASE • VERCEL • SENTRY • POSTHOG

Ronic OS — Revenue Model, Founder Pass & Backend Blueprint

Hybrid Open-Core Strategy, 6-Stream Monetization, Founder Pass Mechanics, PKCE Desktop Auth, Ed25519 Licensing & Cloud Telemetry.

Core App: `Closed-Source Binary` SDKs/Skills: `MIT Open-Source` Pro Margin: `~92%` Gross Founder Cap: `500 Passes ($74.5k)` Offline Lease: `7-Day Ed25519`

1. Open-Core Strategy, 6-Stream Revenue Model & The Founder Pass

Ronic OS follows a **Hybrid Open-Core Model** (like Raycast, Obsidian, and Cursor): the Tauri v2 Rust desktop kernel and licensing engine remain **closed-source** to prevent trivial fork-unlocking, while `@ronic/plugin-sdk`, `SKILL.md` packs, and security audit adapters are published as **MIT Open-Source on GitHub** to drive developer adoption. Because Ronic OS multiplexes the user's own AI subscriptions locally (BYOS), core desktop inference costs you **$0.00/month**:

Tier / Revenue Stream	Pricing (Global / India)	Entitlements & Limits	Creator Unit Economics
1. Ronic Starter	`$0` Free Forever	2 linked models, 1:1 Chat, built-in `Mem0` (500 facts), 1 external bot, 1 machine.	**$0.00** server cost; viral top-of-funnel acquisition loop.
2. Ronic Pro	`$14/mo` or `$120/yr` (`₹999/mo` via UPI)	Unlimited 4-Lane Multiplexer, `<50ms` Quota Failover, Multi-Bot Squads, Auto ⚡ `Supercharge`, 6 Memory Engines, 3 machines.	**~92% gross margin** (all LLM calls run on user's BYOS plans).
3. Founder Pass (LTD)	`$149` One-Time (`₹11,999` • Cap: 500)	Permanent lifetime access to all **Ronic Pro** local desktop features, 3 machine seats, `[★ FOUNDER]` badge, + $10 Burst bonus.	**$74,500 upfront capital** at 500 passes with $0 recurring LLM drain.
4. 24/7 Ghost Runners	`$19/mo` per runner or `$0.008/hr`	Always-on cloud Firecracker microVM in **Bot Fleet** that keeps bots & cron routines running when laptop lid closes.	**~86% gross margin** (~$2.50/mo Hetzner microVM wholesale cost).
5. Burst Compute Credits	Prepaid `$10 / $25 / $50`	Emergency wholesale token routing (DeepSeek V4, Qwen, Bedrock) when all user subscriptions hit their 5-hour cap.	**+20% wholesale markup** deducted from prepaid `compute_wallets`.
6. On-Chain & Store Cut	`0.05%` Builder Rebate + `20%` Store Cut	Hyperliquid Builder Code (`f: 50` = 5 bps) + Jupiter referral on quant trades; 80/20 split on paid Marketplace packs.	**100% passive margin** settled directly on-chain and via Stripe Connect.

2. Cloud Backend Topology (Supabase + Vercel + Sentry + PostHog)

Because Ronic OS executes heavy agent loops, SQLite WAL indexing, and PTY sandboxes locally on the user's workstation, the cloud backend is engineered as an ultra-lean control plane handling **Authentication, Ed25519 License Signing, Multi-Currency Webhooks, Burst Proxy Routing, and Privacy-Scrubbed Telemetry**:

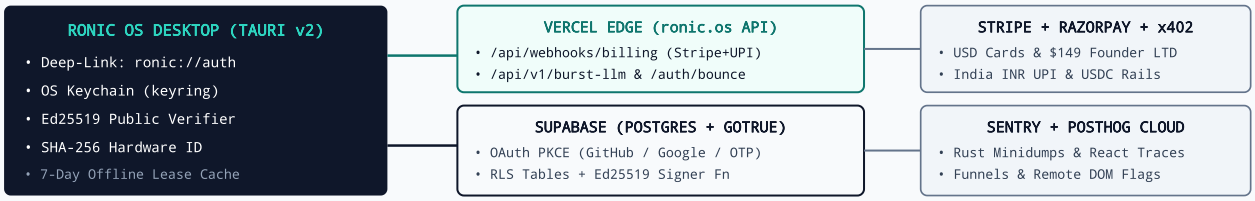


Figure 1: Ronic OS Control Plane — PKCE Deep-Link Auth, Dual Billing Webhooks & Offline-Capable Ed25519 Licensing.

3. Desktop OAuth PKCE & Ed25519 Offline License Lease Protocol

Desktop applications must never rely on a blocking cloud database call on startup—otherwise paying users are locked out on airplanes or unstable Wi-Fi. Ronic OS combines **Supabase PKCE Deep-Linking** with **Cryptographic Offline Leases**:

- **Step 1 — Custom URI Deep-Link Login (`ronic://auth/callback`):** Clicking *"Sign in with GitHub / Google"* launches the user's system browser via Supabase GoTrue PKCE. After authentication, Vercel redirects to `ronic://auth/callback?code=...`, which wakes the Tauri v2 process via `tauri-plugin-deep-link`.
- **Step 2 — OS Keychain Token Persistence & Hardware Seat Binding:** The Rust kernel exchanges the PKCE code for a session, stores the `refresh_token` inside the native OS Keychain (`Windows Credential Manager` / `macOS Keychain` under service `os.ronic.workstation`), and registers a salted SHA-256 hardware fingerprint (`COMPUTERNAME + OS + ARCH`) in `public.device_activations` (enforcing a 3-machine limit for Pro/Founder accounts).
- **Step 3 — Ed25519 Signed 7-Day License Certificate:** A Supabase Edge Function signs a compact JSON lease (`{ user_id, tier, hardware_hash, exp_unix }`) using your private **Ed25519 key**. The Rust binary embeds only the **Ed25519 Public Key** and verifies the certificate locally in `<0.1ms`, granting full offline access for 7 days between silent background refreshes.

> **Why Crackers Cannot Bypass the SQLite State Database**
> Even if a user opens `~/.ronic/state.db` in a SQLite editor and changes `plan_tier` from `free` to `founder_lifetime`, the Rust kernel rejects the change unless the accompanying `ed25519_signature` matches the exact `user_id`, `plan_tier`, and `hardware_hash` signed by your Supabase Edge Function's private key.

4. Production Supabase PostgreSQL Schema & Row-Level Security (RLS)

All cloud state resides in six PostgreSQL tables protected by strict Row-Level Security (RLS). Clients can only `SELECT` their own records; all mutations to licenses, Founder Pass inventory, and wallets require the server-side `SUPABASE_SERVICE_ROLE_KEY`:

Supabase Table	Key Columns & Constraints	RLS Policy & Backend Responsibility
`public.profiles`	`id (UUID FK auth.users)`, `email`, `full_name`, `founder_badge_number`, `referral_code`	Auto-created via Postgres trigger on `auth.users` insert. Tracks user identity and Founder # (1–500).
`public.licenses`	`user_id (UNIQUE)`, `plan_tier` (`free` | `pro` | `founder_lifetime` | `team`), `status`, `max_devices`, `ed25519_lease`	Read-only for authenticated client; updated exclusively by Stripe/Razorpay Vercel webhooks.
`public.founder_pass_pool`	`id (SINGLETON)`, `max_supply (500)`, `claimed_count`, `price_usd (149)`, `price_inr (11999)`	Atomic counter (`claimed_count < 500`) decremented inside a serializable transaction on checkout.
`public.device_activations`	`id`, `user_id`, `hardware_hash (SHA-256)`, `device_name`, `os_platform`, `last_seen_at`	Enforces `COUNT(*) <= licenses.max_devices` (1 for Starter, 3 for Pro & Founder Pass). Users can revoke old laptops.
`public.compute_wallets`	`user_id (PK)`, `balance_micro_usd (BIGINT)`, `auto_reload_enabled`, `updated_at`	Stores prepaid Burst Compute & Ghost Runner credits (`$1.00 = 1,000,000 micro_usd`) with zero floating-point drift.
`public.ghost_runners`	`id`, `user_id`, `bot_handle`, `vm_region`, `container_id`, `status`, `hourly_rate_micro_usd`	Tracks active 24/7 Cloud Ghost Runner Firecracker microVMs and syncs stdout logs via Supabase Realtime.

5. Dual-Gateway Billing Webhooks (Stripe Global USD + Razorpay India UPI)

To maximize conversion across both global developers and Indian builders/quants, Ronic OS runs a unified webhook endpoint on Vercel (`https://ronic.os/api/webhooks/billing`) handling three payment rails:

- **Rail A — Stripe Checkout & Billing Portal (Global USD / EUR / GBP):** Handles recurring `$14/mo` and `$120/yr` subscriptions (`customer.subscription.updated`) and one-time `$149` Founder Pass checkouts (`checkout.session.completed`). Verifies `stripe-signature` before mutating Supabase.
- **Rail B — Razorpay Subscriptions & UPI Intent (India INR):** Native support for instant **UPI QR (GPay, PhonePe, Paytm)**, RuPay, and Indian cards for `₹999/mo` Pro and `₹11,999` Founder Passes without international card failures. Verifies `x-razorpay-signature` via `HMAC-SHA256`.
- **Rail C — Crypto `x402` USDC On-Chain Settlement (Base / Solana):** Allows privacy-focused web3 and quant users to purchase the `$149 USDC` Founder Pass or top up Burst Compute Credits directly from the Ronic Agentic Wallet.

6. Observability, Analytics, Ghost Runners & Deployment Stack

A production desktop OS requires real-time crash visibility, privacy-respecting conversion telemetry, and remote hotfix capabilities so you can update broken WebApp DOM selectors without forcing users to reinstall the binary:

`1 • SENTRY DUAL-SDK CRASH ENGINE`

Rust Kernel + React UI Monitoring

Integrates the `sentry` crate in `src-tauri` (capturing Rust panics, SQLite WAL lock timeouts, and PTY crashes) alongside `@sentry/react` in the UI. A mandatory `beforeSend` hook scrubs all user prompts, file paths, and `sk-*` / `ck_*` keys before leaving the device.

`2 • POSTHOG FUNNELS & REMOTE FLAGS`

Analytics & Zero-Build Hotfixes

Tracks privacy-safe milestone events (`app_booted` , `cli_linked` , `supercharge_triggered` , `founder_pass_clicked`) with `autocapture: false` . Uses **PostHog Remote Feature Flags** to push updated DOM selectors for ChatGPT/Grok WebViews and tune `Harness²` thresholds live.

`3 • 24/7 CLOUD GHOST RUNNERS`

Hetzner / Fly.io MicroVM Orchestrator

When a user toggles `[☁️ Ghost ON]` in **Bot Fleet**, Vercel dispatches a signed job to your worker cluster, spinning up an isolated Firecracker container running their bot's `SOUL.md` and cron schedule while billing `compute_wallets` at `$0.008/hr` or `$19/mo` .

`4 • VERCEL + TAURI AUTO-UPDATER`

Signed Releases & Edge Burst Proxy

Hosts `ronic.os` marketing pages, billing webhooks, and the Tauri v2 auto-update manifest (`/api/releases/latest.json` backed by Cloudflare R2 / GitHub Releases) so Windows (`.msi/.exe`), macOS (`.dmg`), and Linux builds update silently.

7. Production Launch & Environment Configuration Checklist

Required Keys & Go-Live Sequence for Ronic OS v1.0

- **Desktop App (`.env.production` in Tauri):** `VITE_SUPABASE_URL` , `VITE_SUPABASE_ANON_KEY` , `VITE_POSTHOG_KEY` , `VITE_SENTRY_DSN` , and compiled-in `RONIC_ED25519_PUBLIC_KEY` (never bundle service-role or private signing keys inside the desktop binary).
- **Vercel Backend & Supabase Edge Secrets:** `SUPABASE_SERVICE_ROLE_KEY` , `RONIC_ED25519_PRIVATE_KEY` , `STRIPE_SECRET_KEY` , `STRIPE_WEBHOOK_SECRET` , `RAZORPAY_KEY_ID` , `RAZORPAY_WEBHOOK_SECRET` , and `OPENROUTER_WHOLESALE_KEY` (for the 20%-margin Burst Compute Proxy).
- **Day-1 Founder Pass Launch Playbook:** Enable the 500-seat Founder Pass counter on `ronic.os/founders` and inside the Ronic OS header bar (`[★ Claim Founder Pass — $149 Lifetime]`), and trigger your `#omni-growth-squad` bots to publish the launch thread across X, Threads, Instagram, and Hacker News.